CLR as attention: compendium in, gene-association map out

WORKED EXAMPLE (TOY): not results

Input: expression compendium

genes × experiments

toy: 14 x 60; real: thousands of genes, hundreds+ experiments, many labs

published CLR (2007), modified | added here

1 **Dependence** — MI (B-spline) → 2 **CLR** — z-score per gene → 3 **Softmax** — per-gene temp. → 4 **Walk** — t steps + residual

Output: gene x gene map

published CLR | **this work**

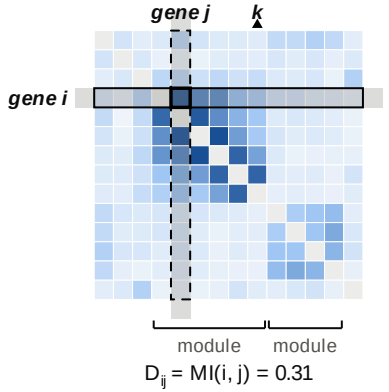
outlines: planted modules (e.g. regulons; not detected)

no learned parameters | unsupervised | expression data only

1 Dependence

MI (B-spline estimator here)

pairwise compatibility (parameter-free, symmetric)

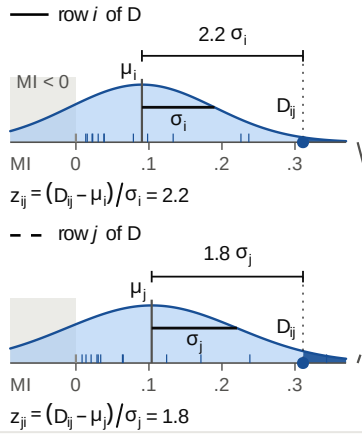


dependence measure: MI (B-spline) here; |r| and normal-scores correlation also tested

2 Per-gene calibration (CLR)

CLR z-scores (Faith 2007); Stouffer combination here

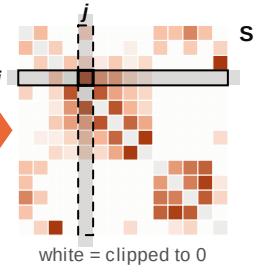
per-query normalization of the logits



$S_{ij} = \frac{z^+_{ij} + z^+_{ji}}{\sqrt{2}}$

$(2.2 + 1.8)/\sqrt{2} = 2.8$

$z^+ = \max(z, 0)$



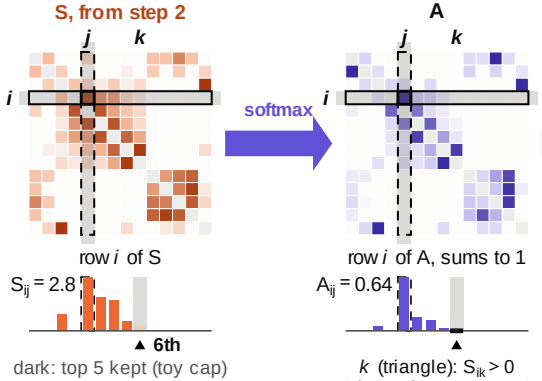
white = clipped to 0

S is the input of step 3

3 Softmax attention

added here

softmax over the keys, temperature per query



$A_i = \text{softmax}(S_i/\tau_i)$; τ_i set so that $k_{eff} = 3$

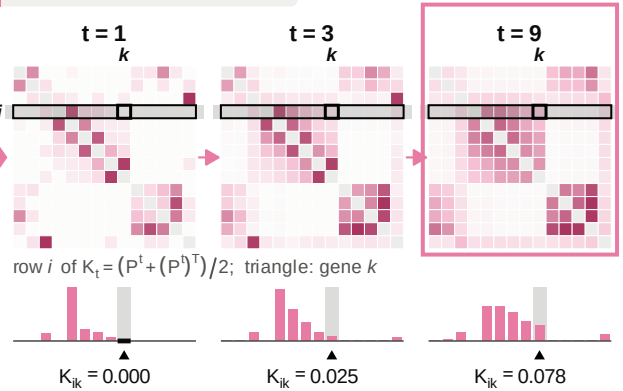
(toy k_{eff} 3, cap 5; real method: k_{eff} 10, top 50)

4 Walk: depth

added here

repeated attention with residual

$P = 0.5I + 0.5A$, repeated t times



toy: average precision (AP) of ranking same-module pairs; favourable by construction. Real data and losses: Figs. 2-6



rand. 0.23 **CLR 0.77**

framed: depth used by the method (chosen without labels on held-out experiments); the toy only displays it

Toy example; analogy with transformer attention, not equivalence; no real-data result shown.